

Google Fiber — Repeat Caller Analysis

How often customers call back after a first inquiry, and why · 65K contacts · 3 markets · Q1 2022

31.2%

Repeat-call rate

64,939

First contacts

44.7%

Worst market (M3)

67.5%

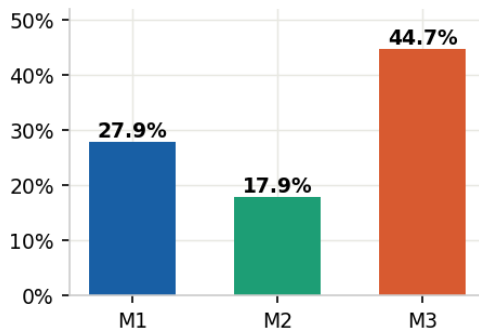
Worst issue (T1)

The situation

Google Fiber's customer-service leadership needs to know how often customers call back after their first inquiry — a direct measure of whether the team resolves issues on first contact. The dashboard breaks repeat-call volume down by market and issue type and tracks it over time, so leadership can see where first-contact resolution is failing and which problems drive the most repeat calls.

What the data shows

Market 3 calls back far more often

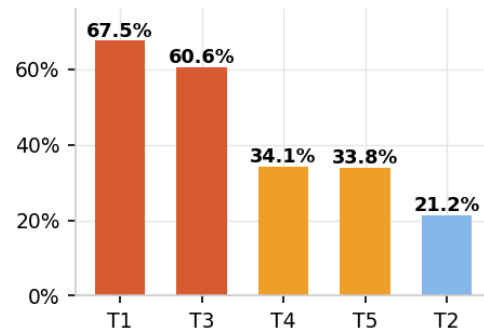


45% of Market 3 first calls return, vs 18% in Market 2 — despite lower volume.

Plus: the overall repeat-call rate is **31.2%** and holds steady week to week — but the single worst hotspot is Market 3 / Type 1, where repeat calls outnumber first calls (a 197% repeat rate).

The dashboard: an interactive workbook. Stakeholders can filter by market and issue type, switch the trend between week / month / quarter / year, and hover any point for exact contact counts and rates.

Issue types 1 & 3 drive repeats



68% of Type 1 and 61% of Type 3 inquiries generate a repeat call.

Recommendations

- **Audit Market 3's first-contact handling first** — it drives the most repeats relative to its volume.
- **Fix the Type 1 / Type 3 playbooks**, the issue categories customers call back about most; target the Market 3 × Type 1 hotspot specifically.
- **Track the 31% repeat rate as a first-contact-resolution KPI** and watch it weekly as changes roll out.

Skills demonstrated

BigQuery SQL (UNION of three market tables) · null-handling & metric definition · repeat-rate calculation · dashboard design · multi-series trend, ranked bar, funnel & heatmap visualisation · time-granularity controls · translating findings into service-operations recommendations.

Data: three Google Fiber market contact tables, unioned in BigQuery into a single reporting table. Full methodology, dashboard, and supporting charts follow on pages 2–7.

01 Data and method

The reporting table unions three market tables (market_1, market_2, market_3) in BigQuery into **1,350 rows representing 64,939 first contacts** across Q1 2022. Each row is a date × issue type × market combination. The columns **contacts_n** (first contacts) and **contacts_n_1 ... contacts_n_7** (1st through 7th repeat contacts) are the heart of the analysis.

The key metric

Repeat-call rate = total repeat contacts (n_1...n_7) ÷ first contacts (n). This answers leadership's question directly: of every inquiry handled, what share generated at least one call-back? A lower rate means better first-contact resolution.

Data-quality checks performed

- **Empty cells = zero:** about 4,200 blank cells across the contact columns represent date/type/market combinations with no contacts of that kind. These were treated as 0, not dropped — confirmed by the cumulative funnel pattern.
- **Granularity & time span:** the data covers one quarter (Q1 2022, ~14 ISO weeks). The dashboard offers week / month / quarter / year views as required; quarter and year are single points until more data accrues, and the writeup notes this rather than implying a multi-year trend.
- **Integrity:** no duplicate rows; three markets and five issue types, balanced 450 rows each.

02 Headline findings

Finding	Evidence
Overall repeat-call rate is 31.2%	20,240 repeat contacts on 64,939 first contacts
Market 3 is the worst performer	44.7% repeat rate, vs 27.9% (M1) and 17.9% (M2)
Issue types 1 and 3 drive repeats	67.5% and 60.6% repeat rate — highest of five types
Volume ≠ repeat problem	M1 has the most contacts but a near-average rate
A clear hotspot exists	Market 3 × Type 1 = 197% (≈2 callbacks per first call)
Rate is stable over the quarter	Overall weekly rate stays ~29–34%

03 Low-fidelity mockup

Before building, I sketched a low-fidelity wireframe to plan the layout: a filter strip and a week/month/quarter/year granularity toggle, a KPI row, a full-width repeat-rate trend by market, and a bottom row of breakdowns — by market, by issue type, and a market × type heatmap.

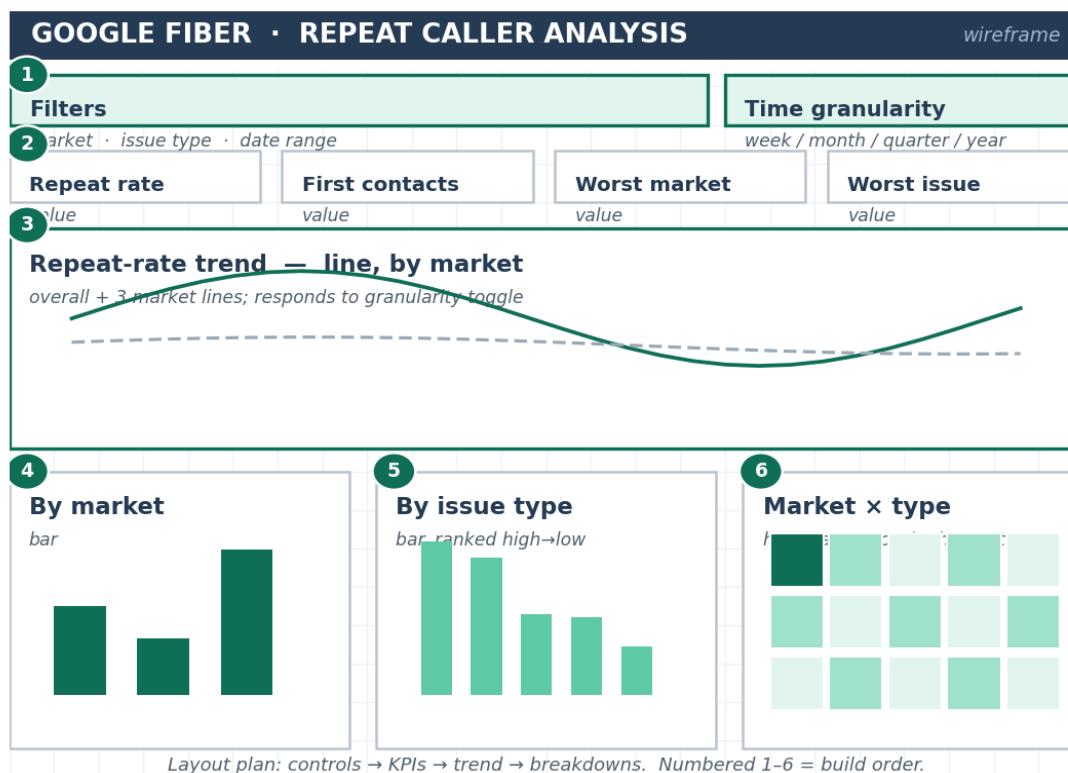


Figure 1 — Low-fidelity wireframe: planned dashboard components and layout.

04 The built dashboard

The wireframe was built out into the interactive workbook below. Filters and the time-granularity toggle sit at the top, KPI cards summarise the headline numbers (including the worst market and issue), and the trend plus four breakdowns answer each stakeholder question.

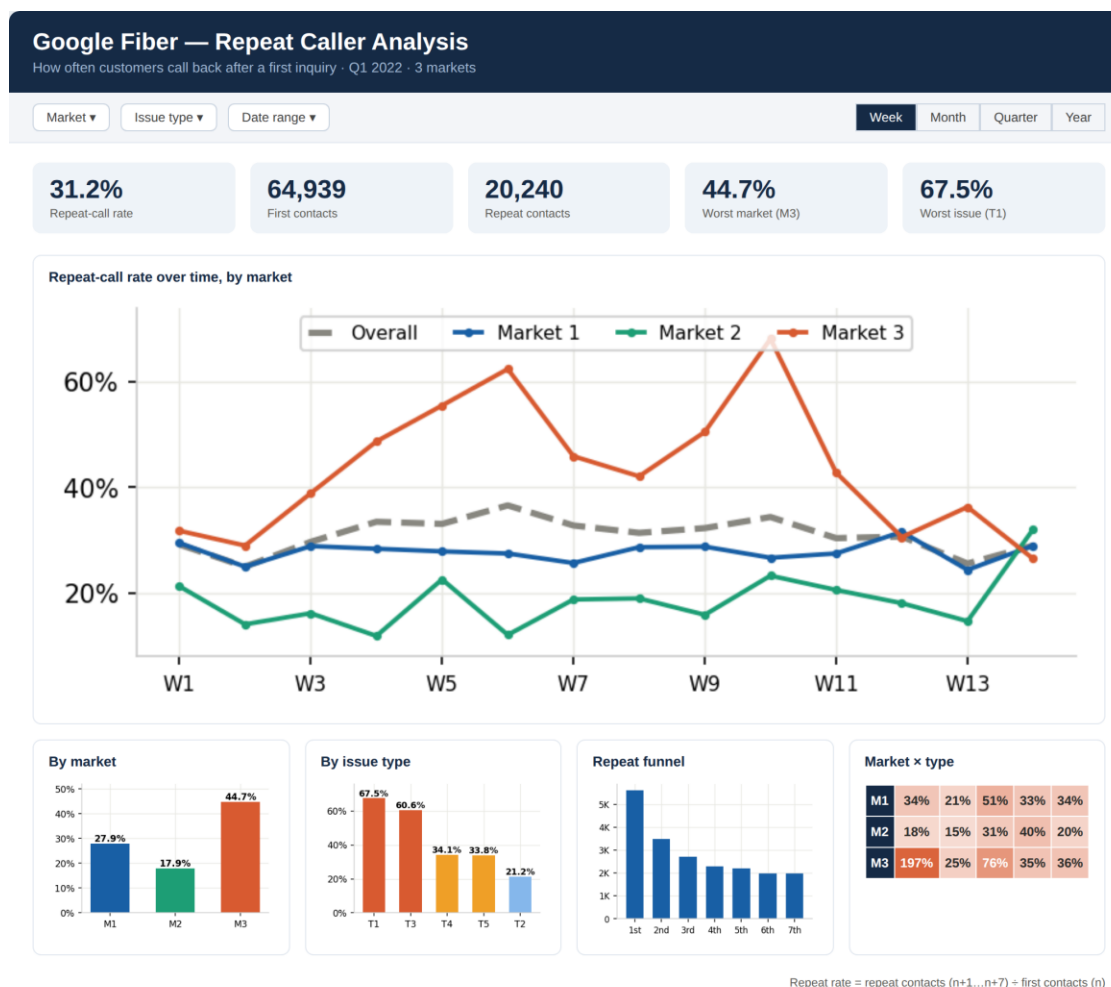


Figure 2 — Snapshot of the built dashboard. Charts update live as filters and the time granularity change.

05 Finding 1 · Repeat rate over time, by market

Tracking the repeat-call rate weekly shows the overall figure holding around 31%, but the markets diverge sharply. **Market 3 (orange) runs far above the others**, spiking past 60% in some weeks, while Market 1 stays near 28% and Market 2 sits lowest around 18%. The week/month/quarter/year toggle lets stakeholders change the time grain; with one quarter of data the weekly view is the most informative.

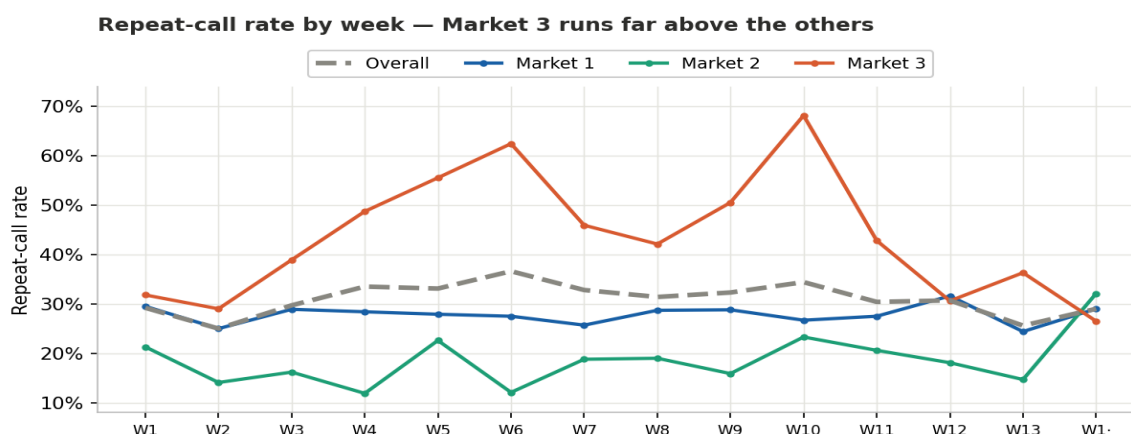


Figure 3 — Repeat-call rate by week, overall and by market.

06 Finding 2 · Which markets and issues drive repeats

By market, the gap is stark: Market 3's 44.7% repeat rate is more than double Market 2's 17.9%. By issue type, **Type 1 (67.5%)** and **Type 3 (60.6%)** generate the most call-backs — roughly two-thirds of those inquiries return — while the high-volume Type 2 has the lowest rate at 21.2%. This tells leadership that the repeat problem is concentrated in specific issue categories, not spread evenly.

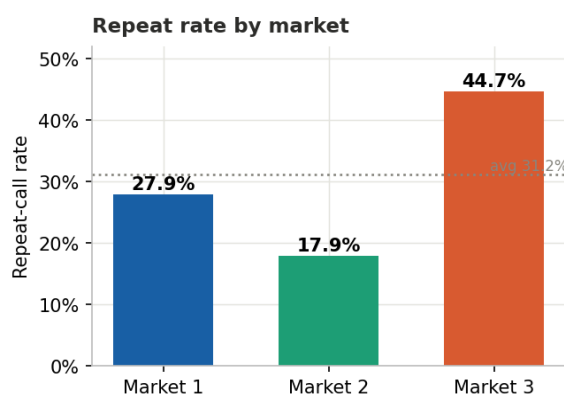


Figure 4 — Repeat rate by market.

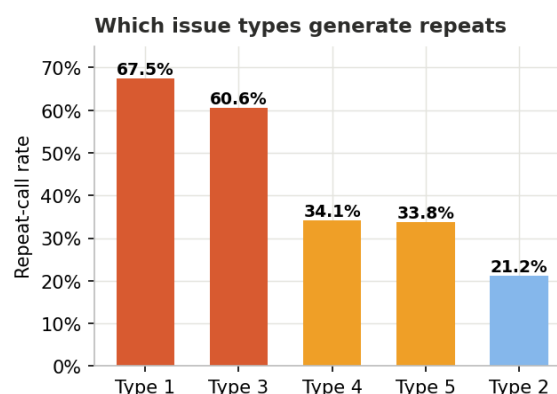


Figure 5 — Repeat rate by issue type.

07 Finding 3 · The repeat funnel

Of first contacts that generate a call-back, the funnel shows how persistent the problem is: **5,601 customers call a first time, and nearly 2,000 are still calling a seventh time**. The slow decay — rather than a sharp drop after the first call-back — signals that a meaningful share of issues stay unresolved across many contacts.

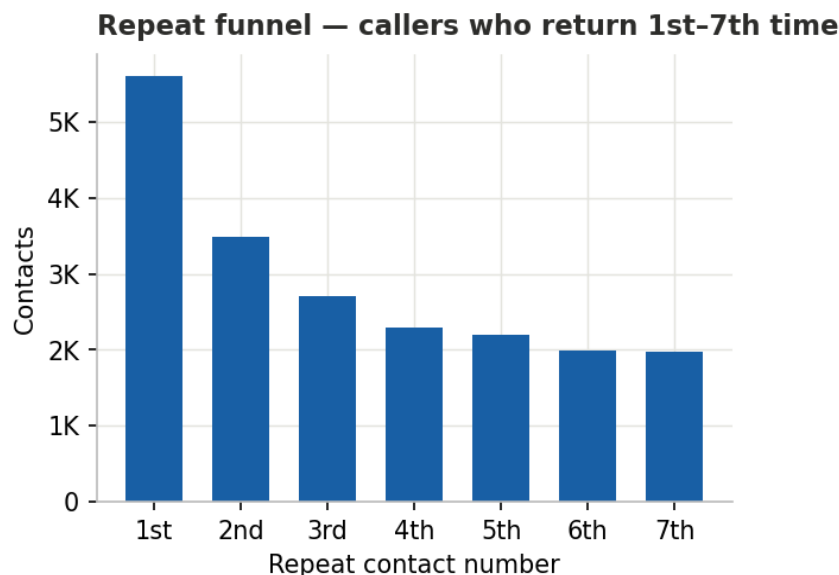


Figure 6 — Repeat funnel: customers returning a 1st through 7th time.

08 Finding 4 · The Market 3 × Type 1 hotspot

Crossing market with issue type pinpoints exactly where to act. Most cells sit in the 15–40% range, but **Market 3 / Type 1 reaches 197%** — those inquiries generate nearly two repeat calls for every first call. Market 3 / Type 3 (76%) is the next worst. This single cell is the most actionable insight in the dashboard.

Repeat rate: market × issue type					
	T1	T2	T3	T4	T5
M1	34%	21%	51%	33%	34%
M2	18%	15%	31%	40%	20%
M3	197%	25%	76%	35%	36%

Figure 7 — Repeat rate by market × issue type; darker = worse.

09 Dashboard capabilities

The workbook is interactive, not a static report. Stakeholders can:

- **Switch the time granularity** between week, month, quarter, and year so trends can be read at any cadence the brief requires.
- **Filter by market and issue type**, redrawing every chart and KPI to isolate, say, just Market 3 or just Type 1.
- **Hover any data point** to read exact first-contact counts, repeat counts, and the repeat rate, rather than estimating from the axis.
- **Read the market × type heatmap** to locate specific problem cells, and the funnel to gauge how persistent repeat calls are.

10 What this means for Google Fiber

- **Investigate Market 3 first.** Its repeat rate is the highest by far; understanding why its first-contact resolution lags is the fastest path to reducing total call volume.
- **Rebuild the Type 1 and Type 3 resolution playbooks.** These issue types come back two-thirds of the time; better first-contact scripts, tooling, or escalation here cut the most repeats.
- **Target the Market 3 × Type 1 hotspot directly**, and adopt the 31% repeat-call rate as a tracked first-contact-resolution KPI reviewed weekly.